

# Chuqin (Allen) Geng

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## Research interests

Neuro-symbolic AI; mechanistic interpretability; interpretable and reliable learning for graph, vision, and language models; neural network robustness and formal verification.

## Education

**University of Toronto** Toronto, ON, CA  
Visiting scholar Jan 2023 – Present  
Advisor: Prof. Xujie Si

**McGill University, Mila - Quebec AI Institute** Montreal, QC, CA  
PhD in Computer Science Sep 2021 – Present  
Advisor: Prof. Xujie Si, GPA: 4.0/4.0

**Georgia Institute of Technology** Atlanta, GA, USA  
MSc in Computer Science, GPA: 4.0/4.0 Sep 2019 – Apr 2021

**University of Reading** Reading, Berkshire, UK  
MSc in Finance, GPA: 4.0/4.0 Sep 2016 – Dec 2017

**University of Toronto** Toronto, ON, CA  
Honours BSc in Math and Statistics Sep 2011 – Aug 2015

## Publications

**Beyond Message Passing: A Symbolic Alternative for Expressive and Interpretable Graph Learning**

**Chuqin Geng**, Li Zhang, Haolin Ye, Ziyu Zhao, Yuhe Jiang, Tara Saba, Xinyu Wang, Xujie Si  
*arXiv preprint, 2026*

**Neural Proposals, Symbolic Guarantees: Neuro-Symbolic Graph Generation with Hard Constraints**

**Chuqin Geng**, Li Zhang, Mark Zhang, Haolin Ye, Ziyu Zhao, Xujie Si  
*arXiv preprint, 2026*

**DreamProver: Evolving Transferable Lemma Libraries via a Wake-Sleep Theorem-Proving Agent**

Youyuan Zhang, Jialiang Sun, Hangrui Bi, **Chuqin Geng**, Wenjie Ma, Zhaoyu Li, Xujie Si  
*COLM 2026*

**LogicXGNN: Grounded Logical Rules for Explaining Graph Neural Networks**

**Chuqin Geng**, Ziyu Zhao, Zhaoyue Wang, Haolin Ye, Yuhe Jiang, Xujie Si  
*ICLR 2026 (Top ~2%)*

**NEUROLOGIC: From Neural Representations to Interpretable Logic Rules**

**Chuqin Geng**, Anqi Xing, Li Zhang, Ziyu Zhao, Yuhe Jiang, Xujie Si  
*UCRL Workshop @ ICLR, 2026*

**Learning Minimal Neural Specifications**

**Chuqin Geng**, Zhaoyue Wang, Haolin Ye, Xujie Si  
*NeuS 2025 (Oral)*

**VisionLogic: From Neuron Activations to Causally Grounded Concept Rules for Vision Models**

**Chuqin Geng**, Yuhe Jiang, Ziyu Zhao, Zhaoyue Wang, Haolin Ye, Anqi Xing, Li Zhang, Xujie Si  
*preprint, 2025*

**Towards Robust Saliency Maps**

Van Nham Le, **Chuqin Geng**, Xujie Si, Arie Gurfinkel  
*ACML 2024*

**TorchProbe: Fuzzing Dynamic Deep Learning Compilers**

Qidong Su, **Chuqin Geng**, Gennady Pekhimenko, Xujie Si  
*accepted to APLAS 2023*

**Scalar Invariant Networks with Zero Bias**

**Chuqin Geng**, Xiaojie Xu, Haolin Ye, Xujie Si  
*accepted to NeurReps @ NeurIPS 2023*

**Towards Reliable Neural Specifications**

**Chuqin Geng**, Nham Le, Xiaojie Xu, Zhaoyue Wang, Arie Gurfinkel, Xujie Si  
*accepted to ICML 2023 (Oral)*

**Identifying Different Student Clusters in Functional Programming Assignments: From Quick Learners to Struggling Students**

**Chuqin Geng**, Wenwen Xu, Yingjie Xu, Brigitte Pientka, Xujie Si  
*accepted to SIGCSE 2023 TS*

## **Novice Type Error Diagnosis with Natural Language Models**

**Chuqin Geng**, Haolin Ye, Yixuan Li, Tianyu Han, Brigitte Pientka, Xujie Si  
*accepted to APLAS 2022*

## **SAT-DIFF: A Tree Diffing Framework Using SAT Solver**

**Chuqin Geng**, Haolin Ye, Yihan Zhang, Brigitte Pientka, Xujie Si  
*preprint, [arxiv.org/abs/2404.04731](https://arxiv.org/abs/2404.04731)*

## **Can ChatGPT Pass An Introductory Level Functional Language Programming Course?**

**Chuqin Geng**, Yihan Zhang, Brigitte Pientka, Xujie Si  
*preprint, [arxiv.org/abs/2305.02230](https://arxiv.org/abs/2305.02230)*

### **Selected awards**

|                                                                   |           |
|-------------------------------------------------------------------|-----------|
| Apple Scholars nomination (McGill University)                     | 2023      |
| Computer Science Top-up Award (McGill University)                 | 2023      |
| Grad Excellence Award (McGill University)                         | 2021–2023 |
| Academic Distinction Award (University of Reading)                | 2017      |
| Second prize in mathematics competition (Shaanxi province, China) | 2009      |

### **Research Experience**

#### **Explainable AI and Mechanistic Interpretability**

Advisor: Prof. Xujie Si (University of Toronto) *May 2025 – Present*  
Developed a novel framework to extract interpretable logic rules and circuits from deep learning models, including GNNs and CNNs.  
Leveraged mechanistic interpretability to diagnose model failure modes and applied symbolic fixes to provide formal guarantees and improve robustness. Extended these neuro-symbolic methods to large language models and agentic theorem proving.

#### **Neural network robustness and verification**

Advisor: Prof. Xujie Si (University of Toronto) *Sep 2022 – May 2025*  
Proposed new specifications for neural network verification. Explored novel methods to improve models' robustness and fairness.

#### **Disproof of a conjecture in biometric security optimization**

Mentors: Prof. Steven Rayan (University of Toronto) *Jan 2015 – Aug 2015*  
Disproved a conjecture regarding optimal solutions for biometric privacy-security trade-offs, providing new bounds for secure system optimization.

### **Teaching experience**

**Head teaching assistant, McGill University** *Winter 2022*  
COMP 302: Programming Languages and Paradigms  
Conducted weekly office hours and tutorials, designed and graded exams, developed auto-graders for assignments, implemented mutation testing, and utilized Moss for plagiarism detection.

## Industry experience

### **FITFI Inc.**

Toronto, CA

Senior Data Scientist

*Jan 2018 – Sep 2019*

Invented the patent “*System and method for automatically detecting and monitoring use of exercise equipment*”. Led algorithm team, secured demo opportunity at 2019 Collision Conference, and helped raise over 2 million CAD funding.

### **SHAREWIN SOFTWARE**

Beijing, China

Algorithm Engineer

*Sep 2015 – Sep 2016*

Designed fault extraction algorithm with preprocessing, filtering, and ant tracking for accurate fault surface detection in 3D-seismic volumes.

## Talks

### **Learning Minimal Neural Specifications**

*May 2025*

NeuS 2025

### **Towards Reliable Neural Specifications**

ICML 2023

*Aug 2023*

### **A study on student performance clusters**

*Mar 2023*

SIGCSE 2023, SPLICE Workshop

### **Novice Type Error Diagnosis with Natural Language Models** *Dec 2022*

APLAS 2022

## Service

I contribute to academic service through conference reviewing and student selection:

- **Machine Learning:** ICML (2023–2026), ICLR (2024–2027), NeurIPS (2023–2026), AAAI (2024–2027)
- **Computer Vision:** CVPR (2024, 2025), ECCV 2026
- **HCI & Education:** SIGCSE 2023, CHI 2023
- **Supervision Evaluation Committee Member:** Mila Supervision Request Process 2027

## Mentoring

Li Zhang, Ziyu Zhao, Haolin Ye, Yihan Zhang, Zhaoyue Wang, Xiaojie Xu